GraamSeva

Name of Institute	Challenge Area	Team Name	Team Lead Email
IIT Roorkee	AI for India Track	Bada Pikachu	stuti_j@cy.iitr.ac.in

1. The Team

GraamSeva started not in a lab, but in a field. During a semester break, Bhavy visited his hometown and while helping his farming families, realised how many welfare schemes — PM-KISAN, crop insurance, soil health subsidies — went unclaimed simply because no one knew about them or could navigate the process. Back on campus, He shared this with us in our college club. When we registered for AI Unlocked, the “AI for India” track was an instant choice — it was personal. We built GraamSeva to do for every farmer across India what we couldn’t do for our own families: make every eligible scheme reachable, in their language, over a simple phone call.

Bhavy — Backend & Azure cloud architecture.

Prachet — API integrations & government portal connectors.

Kabeer — Azure AI Speech NLP & GPT-4o pipeline.

Stuti — Frontend PWA & low-literacy UX design.

2. The Concept

GraamSeva is an AI-powered multi-service platform for rural India, accessible three ways: (1) a missed-call IVR number — zero data, any phone; (2) an offline-first Android PWA; and (3) a web dashboard for CSC operators and ASHA workers. A citizen speaks in their native dialect — Azure AI Speech transcribes in 10 Indian languages. Azure OpenAI (GPT-4o) routes the intent to one of 20+ government service workflows. No English. No smartphone required. No literacy barrier.

The problem we solve is not a technology gap — it is a translation gap. GraamSeva translates between two worlds: the government's English-first digital infrastructure and 700 million citizens who speak Bhojpuri, Awadhi, Odia, and Marathi. By completing the last mile — not just informing but helping them along the way — GraamSeva ensures every eligible rupee reaches every eligible citizen.

3. Target Audience & Market

Dimension	Detail	Source
Geography	Rural & semi-urban India. Phase 1: UP, Bihar, Odisha, Maharashtra — highest scheme density + lowest digital literacy	MoRD State Rankings, 2023
Demographics	Adults 18–65 on any mobile phone. Focus: women (do 60–80% farm work but hold <13% land) and landless labourers	ICAR Gender Report, 2022
Platform	Missed-call IVR + WhatsApp+ Android PWA. No smartphone required for core service	TRAI & Meta India Reports

Market Size	180M MGNREGA card holders · 110M PM-KISAN beneficiaries · 800M ration card holders = 1B+ addressable touchpoints	MoRD Annual Report, 2023–24
Channels	Common Service Centres (5.5L CSCs) · KVKs · ASHA workers · Anganwadi — trusted last-mile partners already present	Min. of Electronics & IT, 2024

4. Personas

Ramkali Devi, 47

Sitapur, UP · Farmer · Keypad phone

"Mujhe PM-KISAN ka paisa milna chahiye, par form samajh nahi aata."

Pain: Denied PM-KISAN twice; travels 12km to CSC; no smartphone

Need: Hindi voice claim filing + IVR callback on approval

Suresh Nayak, 32

Koraput, Odisha · MGNREGA labourer · No data

"Hamare gaon mein koi nahi batata ki hum kisi cheez ke haqdar hain."

Pain: Speaks only Odia tribal dialect; CSC 20km away, open 3 days/week

Need: Odia voice MGNREGA registration + zero-data missed-call access

5. How It Works — Architecture

GraamSeva runs on a five-stage voice-first pipeline:

Citizen Voice Input	Azure AI Speech Dialect STT	Azure OpenAI Intent Router	Gov APIs	IVR/SMS Voice Callback

Azure services powering GraamSeva:

Service	Role	Why India-specific
Azure AI Speech	Dialect STT + TTS, 10 Indian languages	Covers Bhojpuri, Awadhi, Odia dialects — not just Hindi
Azure OpenAI GPT-4o	Intent routing, eligibility logic, form generation	Handles complex multi-step scheme logic in natural Hindi/vernacular
Azure IoT Edge	Offline inference node — 20 most common queries cached	Ensures 2G continuity; syncs to cloud when connectivity resumes
Azure Maps	Mandi prices, cold storage, APMC locations	Field-level precision vs national-level govt weather/price data
Azure Monitor	Real-time impact dashboard: calls handled, ₹ unlocked	Transparent proof of impact for govt partners and CSR funders

6. Core Technologies

Platform / Technology	Role
Microsoft Azure AI Speech (STT + TTS)	Dialect-aware voice I/O — the access layer that makes everything else possible
Azure OpenAI (GPT-4o)	Central reasoning: scheme eligibility, form generation, multi-turn conversation in 10 languages
Azure IoT Edge (offline inference)	2G-resilient pipeline — 48hr local buffer, sync on reconnect
IVRS backbone (Exotel + Azure integration)	Missed-call queuing, IVR flow, outbound callback — 99.9% uptime, trusted Indian infra
Android PWA (React Native offline-first)	App that works offline, syncs on Wi-Fi, <5MB install, works on Android 8+
Azure Maps + APMC Price API	Real-time mandi prices + field-level weather at village resolution
Azure Monitor + Application Insights	Live impact dashboard — calls handled, schemes filed, ₹ unlocked, geographic heatmap

7. The Business Plan

GraamSeva follows a district-first go-to-market strategy — starting deep in a few high-impact districts, proving measurable outcomes, and then scaling outward. Revenue comes from government contracts and CSR partnerships, never from the farmers we serve. Below is our phased rollout and revenue plan.

Phase-wise Rollout

Phase	Target Districts	Goal	Revenue Trigger
Phase 1 — Pilot (Months 1–4)	Lucknow & Sitapur (UP), Patna & Muzaffarpur (Bihar) — highest MGNREGA + PM-KISAN density	5,000 calls handled · 1,000 scheme applications filed · Onboard 2 local NGO partners · Establish CSC tie-ups	CSR pilot grants from local corporates; impact data used as proof-of-concept for govt proposal
Phase 2 — District Expansion (Months 5–12)	+10 districts across UP, Bihar & Odisha (Koraput, Balangir, Gaya, Varanasi, Gorakhpur, Darbhanga, Samastipur)	50,000 MAUs · 15 languages live · Livestock & crop insurance modules added · 1 state govt MoU signed	Per-application fee billed to state govt (B2G); CSR contracts with 2–3 corporates operating in target districts
Phase 3 — State Scale (Year 2+)	All districts of UP, Bihar & Odisha; expand into Maharashtra & MP — targeting 8 states, 200+ districts	10M users · Official Digital India partner · API licensing to rural fintech & insurers · Dialect NLP open-sourced	Multi-state B2G contracts · API licensing (B2B) to NGOs, rural fintech, PMFBY insurers · Anonymised data insights to policy bodies

8. Additional Information

Research & Validation:

- Conducted 15 informal interviews with individuals from rural UP, Bihar and Odisha backgrounds to validate friction points in accessing government services.
- Tested voice comprehension of Bhojpuri and Maithili inputs against Azure AI Speech SDK — 87% word accuracy at 2G audio quality.
- All scheme data cross-referenced against MyScheme.gov.in (official GoI scheme portal) and individual ministry dashboards.

Impact Metrics (Azure Monitor Dashboard):

- Total calls handled per day / per district
- Successful applications filed + estimated ₹ value unlocked per applicant
- End-to-end time: call received → outcome confirmed (target <5 minutes)
- Language distribution of callers (diversity signal)
- Offline vs online session ratio (connectivity signal)

Open-Source Commitment:

The government API connector layer (state portal abstraction) will be open-sourced post-hackathon to let other teams build on India's fragmented public digital infrastructure.

Submission Details

The All-in-One AI Helper for Rural India — Voice + App + Web

Bhavy · Prachet · Kabeer · Stuti | IIT Roorkee · Batch of 2028 · AI for India Track